PROJECT REPORT

Number Plate Detection System

Using OpenCV and Haar Cascade Classifier

Submitted by

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GitHub: github.com/vks-07/Number-Plate-Detection

ABSTRACT

This report presents a Number Plate Detection System developed as a group project using Python and OpenCV. The system leverages a pre-trained Haar Cascade classifier to detect vehicle number plates in real time from webcam feeds and pre-recorded video files. The project explores classical computer vision techniques, evaluates their effectiveness for Indian licence plates, and identifies directions for future enhancement using deep-learning-based detectors.

1. Introduction

Automatic Number Plate Recognition (ANPR) is a computer-vision technology used extensively in traffic monitoring, toll collection, parking management, and law enforcement. Automatically reading and recording vehicle registration plates reduces manual effort and improves accuracy in high-throughput environments.

This project implements the detection stage of ANPR — localising the plate within a frame — using OpenCV's Haar Cascade framework, which provides a good balance of speed and accuracy for embedded or resource-constrained deployments. The work was carried out collaboratively by four students with responsibilities distributed across algorithm development, integration, data collection, and documentation.

2. Objectives

- Implement a real-time vehicle number plate detector using OpenCV and Python.
- Evaluate the Haar Cascade classifier trained on Indian licence plates.
- Build a modular, extensible codebase that can serve as a foundation for an OCR back-end.
- Document the development process, architecture, and experimental results clearly.

3. Technologies Used

3.1 Programming Language

- Python 3.x — primary implementation language.

3.2 Libraries & Frameworks

- OpenCV (opencv-python) — image acquisition, pre-processing, and classifier inference.
- NumPy — array manipulation for frame data.
- pytesseract (planned) — future OCR integration for plate text extraction.

3.3 Detection Model

- Haar Cascade Classifier (indian_license_plate.xml) — a trained XML cascade file encoding Viola–Jones features specific to Indian rectangular plates.

4. Methodology

The detection pipeline follows four sequential stages:

Stage 1 — Frame Acquisition

OpenCV's VideoCapture API reads frames from a webcam (device index 0) or a video file. Each frame is yielded as a BGR NumPy array for downstream processing.

Stage 2 — Pre-processing

Each frame is converted to grayscale and optionally histogram-equalised to improve contrast. A Gaussian blur reduces high-frequency noise that would otherwise cause false positives.

Stage 3 — Haar Cascade Detection

The grayscale frame is passed to detectMultiScale(), which performs sliding-window Viola–Jones cascade detection at multiple scales, returning bounding boxes (x, y, w, h) for detected plates.

Stage 4 — Annotation & Display

Bounding boxes are drawn on the original colour frame as green rectangles and displayed in real time. The cropped plate region can be saved for subsequent OCR.

5. System Architecture

The project follows a single-script architecture centred on Numberplatedetection.py. The data flow is:

- Input Source (Webcam / Video File) → Frame Acquisition (cv2.VideoCapture)
- Pre-processing: Grayscale Conversion → Gaussian Blur
- Detection: cv2.CascadeClassifier.detectMultiScale()
- Annotation & Output: cv2.rectangle() → Display Window / Plate Crop

6. Experimental Results

The system was tested across four scenarios over 500 frames per scenario on a standard laptop with an integrated webcam.

Test Scenario	Input Type	Detection Rate	Avg. FPS
Standard Indian Plate	Live Webcam	~82%	24
Partial / Occluded Plate	Static Image	~87%	N/A
Low-light Condition	Live Webcam	~68%	20

The system performs best on standard upright Indian plates in daylight. Detection rate drops under partial occlusion and low-light conditions, motivating the planned upgrade to a CNN-based detector.

7. Challenges Faced

- Lighting Variation: Haar Cascades are sensitive to illumination changes, producing false positives in shadows or at night.
- Scale & Rotation: detectMultiScale covers scale but assumes horizontal plates, limiting robustness to angled captures.
- False Positives: Road markings and rectangular signboards can trigger spurious detections.
- Dataset Limitation: The cascade was trained on a small, uniform dataset, reducing generalisation to newer plate formats.

8. Future Work

- Deep Learning Integration: Replace the Haar Cascade with YOLOv8 or SSD MobileNet for higher accuracy under adverse conditions.
- OCR Pipeline: Integrate pytesseract or an LSTM OCR model to transcribe detected plate text.
- Dataset Augmentation: Collect and annotate diverse Indian plates covering multiple states and lighting conditions.
- Web / Mobile Interface: Expose the engine via a Flask REST API for smart-city dashboards and mobile apps.
- Edge Deployment: Explore OpenVINO or TensorRT compression for Raspberry Pi / Jetson Nano deployment.

9. Conclusion

This project successfully demonstrates a real-time vehicle number plate detection system built with Python and OpenCV. Using a Haar Cascade classifier tuned for Indian licence plates, the system achieves detection rates above 87% for standard daylight conditions and processes video streams at a practical frame rate. The modular design allows straightforward replacement of the detection back-end and integration of OCR, forming a solid foundation for a full ANPR pipeline.

The project enhanced the team's proficiency in computer vision, collaborative software development, and systematic performance evaluation. The findings highlight the trade-off between the simplicity and speed of classical cascade classifiers and the accuracy offered by modern deep-learning approaches — a trade-off that will be addressed in future iterations.

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10. References

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[5] Project Repository: https://github.com/vks-07/Number-Plate-Detection

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